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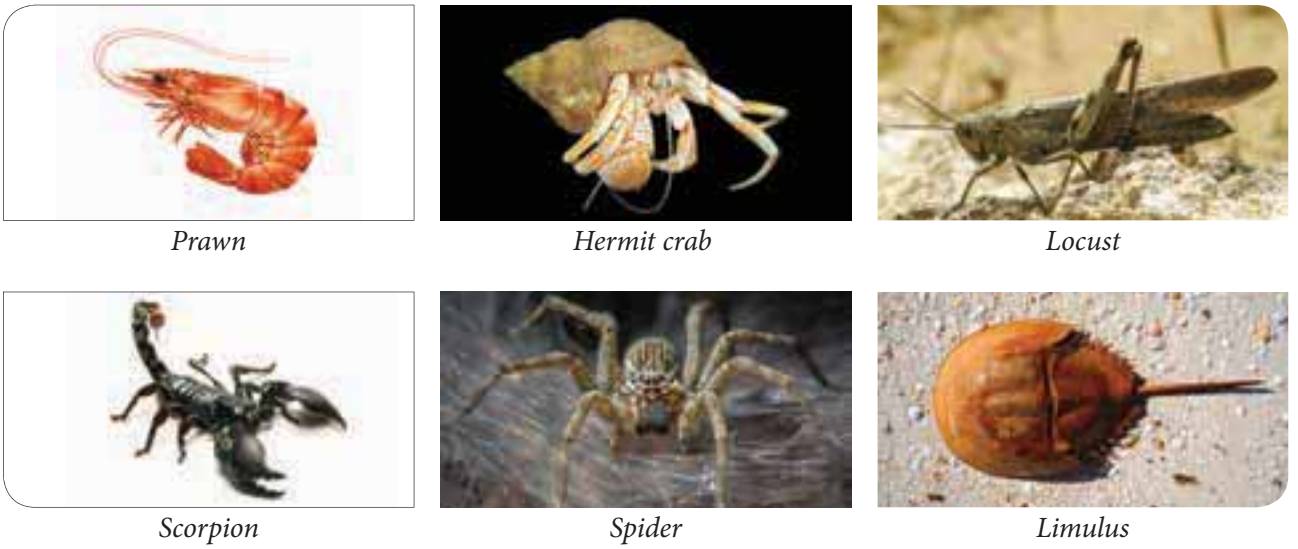


Figure 2.15 Examples of Arthropoda

feeding and are sensory in function. Body is covered by chitinous exoskeleton for protection and to prevent water loss, It is shed off periodically by a process called moulting or ecdysis. The body consists of a head, thorax, and abdomen with a body cavity called haemocoel. Respiratory organs are gills, book gills, book lungs or trachea. Circulatory system is of open type. Sensory organs like antennae, eyes (compound and simple), statocysts (organs of balance/ equilibrium) are present. Excretion takes place through malpighian tubules, green glands, coxal glands, etc. They are mostly dioecious and oviparous; fertilization is usually internal. Development may be direct or indirect. Life history includes many larval stages followed by metamorphosis.

Examples : *Limulus* (King crab, a living fossil), *Palamnaeus* (Scorpion), *Eupagarus* (Hermit crab), *Apis* (Honey bee), *Musca* (House fly), Vectors- *Anopheles*, *Culex*, *Aedes* (mosquitoes), Economically important insects - *Apis*- (Honey bee), *Bombyx* (Silk worm), *Laccifer* (Lac insects), Living fossils *Limulus*- (King crab), Gregarious pest - *Locusta* (Locust) (Figure 2.15)

2.3.8 Phylum: Mollusca

(L. *molluscs* –soft bodied)

This is the second largest animal phylum. Molluscs are terrestrial or aquatic (marine or fresh water) and exhibit organ system level of body organisation. They are bilaterally symmetrical (except univalves), triploblastic and coelomate animals. Body is covered by a calcareous shell and is unsegmented with a distinct head, muscular foot and a visceral hump or visceral mass. A soft layer of skin forms a mantle over the visceral hump. The space between the visceral mass and mantle (pallium) is called the mantle cavity in which a number of feather like gills (**ctenidia)** are present, which are respiratory in function. The

Spider silk is five times stronger than steel of the same diameter. It has been suggested that a Boeing 747 could be stopped in flight by a single pencil-width strand and spider silk is almost as strong as Kevlar, the toughest man-made polymer.